Capacitors — Variables Reference

Capacitance, charging/discharging and AC behaviour: names, symbols, units and the formulas each quantity appears in.

Name	Symbol	Unit	Unit symbol	Common formula(s)
Capacitance	**C**	farad	**F**	$C = Q \div V$ $\tau = RC$ $C = \varepsilon_0\varepsilon_r A \div d$ $Xc = 1 \div (2\pi fC)$ Series: $1 \div CT = 1/C1 + 1/C2 + \ldots$ Parallel: $CT = C1 + C2 + C3 + \ldots$
Charge	**Q**	coulomb	**C**	$Q = CV$ $E = \frac{1}{2}QV$
Voltage	**V**	volt	**V**	$Q = CV$ $E = \frac{1}{2}QV$
Energy stored	**E**	joule	**J**	$E = \frac{1}{2}QV$ $E = \frac{1}{2}CV^2$
Time constant	**τ (tau)**	second	**s**	$\tau = RC$
Capacitive reactance	**Xc**	ohm	**Ω**	$Xc = 1 \div (2\pi fC)$
Impedance	**Z**	ohm	**Ω**	$Z = \sqrt{(R^2 + Xc^2)}$
Dielectric constant / relative permittivity	**ε_r (epsilon sub r)**	(no unit — a ratio)	**—**	$C = \varepsilon_0\varepsilon_r A \div d$
Plate area	**A**	square metre	**m^2**	$C = \varepsilon_0\varepsilon_r A \div d$
Plate separation	**d**	metre	**m**	$C = \varepsilon_0\varepsilon_r A \div d$
Frequency	**f**	hertz	**Hz**	$Xc = 1 \div (2\pi fC)$
Permittivity of free space	**ε_0 (epsilon nought)**	farad per metre	**$F\ m^{-1}$**	Constant: $\varepsilon_0 = 8.85 \times 10^{-12}\ F\ m^{-1}$ $C = \varepsilon_0\varepsilon_r A \div d$